Chapter 5 solutions

CH05-E1

1. Cofactors:

$$C_{11} = 1 - 3 = -2,$$
$$C_{12} = -(0 - 6) = 6.$$

The third entry of row 1 is zero, so

$$\det A = 1(-2) + 2(6) = 10.$$

2. Adding twice column 1 to column 2 gives

$$B = \begin{bmatrix} 1 & 4 & 0 \\ 0 & 1 & 3 \\ 2 & 5 & 1 \end{bmatrix}.$$

Expansion along row 1 gives

$$\det B = 1(1 - 15) - 4(0 - 6) = 10.$$

3. Write $B = AE$, where the elementary column-addition matrix satisfies $\det E = 1$. Therefore

$$\det B = \det A \det E = \det A.$$

Geometrically, the shear preserves area or volume.

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CH05-E2

1. The six terms are

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$
$$- a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

2. Column orders

$$(1, 2, 3), \quad (2, 3, 1), \quad (3, 1, 2)$$

are even. The other three are odd.

3. Take

$$M = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 4 \end{bmatrix}.$$

Only

$$2 \cdot 3 \cdot 4 = 24$$

and

$$1 \cdot 1 \cdot 1 = 1$$

Chapter 5 solutions (continued)

are nonzero. Both have even sign, giving

$$\det M = 25.$$

CH05-E3

1.
$$C_{11} = 10, \qquad C_{12} = 4, \qquad C_{13} = -2.$$

2.
$$\det A = 2(10) + 1(4) + 0(-2) = 24.$$

3. The row 2 cofactors are
$$C_{21} = -(4) = -4,$$
$$C_{22} = 8, \qquad C_{23} = -(4) = -4.$$

Hence

$$\det A = (-1)(-4) + 3(8) + 1(-4) = 24.$$

CH05-E4

1. The determinant is 10, and the cofactor matrix is
$$C = \begin{bmatrix} -2 & 6 & -2 \\ -2 & 1 & 3 \\ 6 & -3 & 1 \end{bmatrix}.$$
$$\operatorname{adj} A = C^T = \begin{bmatrix} -2 & -2 & 6 \\ 6 & 1 & -3 \\ -2 & 3 & 1 \end{bmatrix}.$$
$$A^{-1} = \frac{1}{10} \begin{bmatrix} -2 & -2 & 6 \\ 6 & 1 & -3 \\ -2 & 3 & 1 \end{bmatrix}.$$

2. Multiplication gives diagonal entries 10 and off-diagonal entries 0.
$$A \operatorname{adj} A = 10I.$$

3. For $Ax = b$,
$$x = \frac{1}{\det A} \operatorname{adj}(A) b.$$

This is exact but expensive at large size. The adjugate has n^2 minors; use pivoted factorization for large numerical systems.

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Chapter 5 solutions (continued)

CH05-E5

1.

$$\det A = -2, \qquad \det A_1 = -4, \qquad \det A_2 = 2.$$

$$x_1 = 2, \qquad x_2 = -1.$$

2. Verification:

$$3(2) + 4(-1) = 2, \qquad 5(2) + 6(-1) = 4.$$

3. For $b = (2, 5)^T$,

$$\det A_1 = -8, \qquad \det A_2 = 5,$$

$$x_1 = 4, \qquad x_2 = -\frac{5}{2}.$$

CH05-E6

1.

$$P = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \qquad \det P = 5.$$

The counterclockwise ordering gives positive orientation, so signed and unsigned areas are both 5.

2. Define

$$X = w_2 z_3 - w_3 z_2,$$

$$Y = w_3 z_1 - w_1 z_3, \qquad Z = w_1 z_2 - w_2 z_1.$$

Direct expansion gives

$$w_1 X + w_2 Y + w_3 Z = 0,$$

because the terms cancel in pairs. The same expansion proves $z \cdot (w \times z) = 0$.

3.

$$w \times z = (-2, 1, 1)^T.$$

Triple product identity:

$$\det[w \; z \; w \times z] = \|w \times z\|^2 = 6.$$

CH05-E7

1. Factor c from each of the n columns.

$$\det(cA) = c^n \det A.$$

2.

Chapter 5 solutions (continued)

$$1 = \det(Q^T Q) = (\det Q)^2,$$

Hence

$$\det Q = \pm 1, \qquad |\det Q| = 1.$$

3. If $S^T = -S$ and n is odd,

$$\det S = \det(S^T) = \det(-S) = (-1)^n \det S = -\det S.$$

Thus $\det S = 0$, proving singularity.

If n is even, the sign is positive and this argument imposes no zero determinant. For example,

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

has determinant 1.

CH05-E8

1. Expanding along row 1 gives

$$M_{11} = \begin{bmatrix} d & 1 \\ 2 & a+d \end{bmatrix},$$

$$\det M_{11} = d(a+d) - 2.$$

$$M_{12} = \begin{bmatrix} c & 1 \\ 0 & a+d \end{bmatrix},$$

$$\det M_{12} = c(a+d).$$

Since the third entry is zero,

$$\det A = a[d(a+d) - 2] - bc(a+d).$$

2. Choose $a = d = 0, b = c = 1$, giving

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}.$$

Columns 1 and 3 both equal $(0, 1, 0)^T$, so rank is below 3 and the determinant is zero. The columns lie in a plane, and their three-dimensional volume vanishes.

3. $a = d = 1, b = c = 0$:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix}.$$

$$\det A = 1(2 - 2) = 0.$$

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Chapter 5 solutions (continued)

The trace is $1 + 1 + 2 = 4$. The eigenvalue product is the zero determinant, so at least one eigenvalue is zero, agreeing with deficient rank.

Verification code appendix: determinants.py

```
"""Recompute the numerical checks in the chapter five notes."""

from itertools import permutations
from math import isclose


Matrix = tuple[tuple[float, ...], ...]


def determinant(matrix: Matrix) -> float:
    n = len(matrix)
    if n == 0:
        return 1.0
    total = 0.0
    for permutation in permutations(range(n)):
        inversions = sum(
            permutation[i] > permutation[j]
            for i in range(n)
            for j in range(i + 1, n)
        )
        product = 1.0
        for row, column in enumerate(permutation):
            product *= matrix[row][column]
        total += (-1 if inversions % 2 else 1) * product
    return total
```

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Chapter 5 solutions (continued)

```
def minor(matrix: Matrix, row: int, column: int) -> Matrix:
    return tuple(
        tuple(value for j, value in enumerate(values) if j != column)
        for i, values in enumerate(matrix)
        if i != row
    )


def cofactors(matrix: Matrix) -> Matrix:
    return tuple(
        tuple(
            (-1) ** (i + j) * determinant(minor(matrix, i, j))
            for j in range(len(matrix))
        )
        for i in range(len(matrix))
    )


def transpose(matrix: Matrix) -> Matrix:
    return tuple(zip(*matrix))


def multiply(left: Matrix, right: Matrix) -> Matrix:
    return tuple(
        tuple(
```

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Chapter 5 solutions (continued)

```
            sum(left[i][k] * right[k][j] for k in range(len(right)))
            for j in range(len(right[0]))
        )
        for i in range(len(left))
    )


def cramer(matrix: Matrix, right: tuple[float, ...]) -> tuple[float, ...]:
    base = determinant(matrix)
    if isclose(base, 0.0, abs_tol=1e-12):
        raise ValueError("Cramer's rule needs a nonsingular matrix")
    answers = []
    for column in range(len(matrix)):
        changed = [list(row) for row in matrix]
        for row in range(len(matrix)):
            changed[row][column] = right[row]
        answers.append(determinant(tuple(tuple(row) for row in changed)) / base)
    return tuple(answers)


a = (
    (1.0, 2.0, 0.0),
    (0.0, 1.0, 3.0),
    (2.0, 1.0, 1.0),
)
assert isclose(determinant(a), 10.0)

cofactor_matrix = cofactors(a)
adjugate = transpose(cofactor_matrix)
assert multiply(a, adjugate) == tuple(
    tuple(10.0 if i == j else 0.0 for j in range(3))
    for i in range(3)
)

b = (1.0, 2.0, 3.0)
solution = cramer(
    ((3.0, 4.0), (5.0, 6.0)),
    (2.0, 4.0),
)
assert solution == (2.0, -1.0)

u = (2.0, 1.0)
v = (1.0, 3.0)
signed_area = u[0] * v[1] - u[1] * v[0]
assert signed_area == 5.0

print("det(A) =", determinant(a))
print("A adj(A) =", multiply(a, adjugate))
print("Cramer solution =", solution)
print("signed area =", signed_area)
```

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Chapter 5 solutions (continued)

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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